

Module	Topic to Cover	Details/Key Concepts	Duration in Hours
JavaScript Fundamentals	JS Setup & Basics	Node.js, npm, VS Code setup. Variables (var, let, const), Data Types, Operators.	1
	Functions & Scope	Function declaration vs. expressions, Arrow functions, scope (global, function, block).	1
	Control Flow & Loops	Conditional Statements (if/else, switch), Iteration (for, while, forEach).	1
	Modern JS (ES6+)	Promises, Async/Await (Crucial for Playwright), Template Literals, Destructuring, Modules (import/export).	2
Playwright UI Automation & Basics	Playwright Introduction	Architecture (Out-of-Process), Advantages (Auto-wait, Cross-Browser), Installation.	1
	Playwright Tooling	Test Runner, Codegen, Playwright Inspector, Trace Viewer (Debugging).	2
	Locators & Assertions	Web-First Locators (getByRole, getByText), CSS/XPath best practices, built-in expect assertions.	2
	Basic UI Interactions	Navigation (goto), Typing (fill), Clicking, Dropdowns, Checkboxes/Radio buttons.	1.5
	Advanced UI Handling	Handling Iframes, Dialogs (Alerts/Prompts), File Uploads/Downloads, Dynamic waits, Shadow DOM (optional deep-dive).	2.5
	Context & Fixtures	Browser Contexts (Isolation, Authentication State), Using Playwright Fixtures (page, test).	1
Framework Design & Best Practices	Framework Design Principles	Modularity, reusability, maintainability. Setting up project structure (tests, pages, utils).	1
	Page Object Model (POM)	Concepts: Encapsulating locators/actions. Implementation: Page Classes, using Page Fixtures.	3
	Behavior-Driven Development (BDD)	Gherkin Syntax (Given, When, Then), Cucumber/Playwright integration, Feature Files, Step Definitions.	3
	Data-Driven Testing (DDT)	Reading test data from JSON/CSV files, parameterizing tests.	1.5
	Utility & Helper Classes	Creating reusable functions (e.g., random data generation), handling environment variables (.env).	1.5
	Test Configuration	Deep dive into playwright.config.js: Projects, Global Setup/Teardown, Timeouts, Screenshots, Videos, Retries.	2
API, DB, Execution & Reporting	API Automation	Playwright's request fixture for RESTful API testing (GET, POST, PUT, PATCH, DELETE).	2
	API Test Implementation	Making requests, validating status codes, headers, and JSON body/schema. Authentication.	2
	DB Connection and Testing	Connecting to a database (using Node.js libraries). Data setup and teardown via DB queries.	2
	Parallel Execution	Understanding and configuring Playwright's native parallel test execution (workers in config.js).	2
	Reporting	Playwright's built-in HTML Reporter. Integrating and configuring advanced reports like Allure Reporter.	4
Git & CI/CD Deployment	Version Control (Git)	Git Basics: Committing, Branching, Merging. Pushing test code to a remote repository (GitHub/GitLab).	2
	CI/CD Pipeline Creation	CI/CD Concepts (Build, Test, Deploy). Introduction to a CI Tool (e.g., GitHub Actions, Jenkins).	3
	CI/CD Setup & Integration	Setting up the CI/CD pipeline (YAML/Jenkinsfile) to pull test code and execute the Playwright suite.	3

		Configuring the pipeline to automatically run the full test suite after a successful application code deployment (using webhooks/triggers).	3
	Auto-Execution on Deployment		
		Total	50 Hours

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